By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

Macaulay Beam Analysis - BetaTesting

13 - Steel - Single Span -Moment

Bridge, L =		in	E =	29,000,000	lb / in^2	Length Units	in
Beam Sect =	W14x61	W14x90	G =	1.740E+06	lb / in^2 = 0.06*29,000,000	Force Units	lb
A =	17.90	26.50 in^2					
I =	640.00	999.00 in^4					

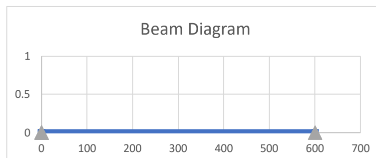
Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in^2	in^4	lb / in^2	lb / in^2
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

	Supp. No.	X	Imposed Movement * dY (vert)	Imposed Movement ** dr (rot)
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	0.0	0.000
	3			
	4			
	5			

* (settlement) ** (Not Working)



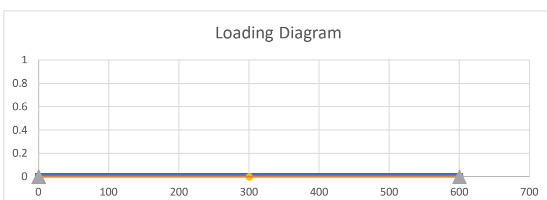
Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)	Load
		in	in	lb / in	lb / in	lb / ft
DO NOT DELETE ROWS (GRAPHED)	1	0.0	600.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0	0.0
	5					

Point Loads - UNFACTORED

	Load No.	X	Py	My	Load
		in	lb	lb-in	lb-ft
DO NOT DELETE ROWS (GRAPHED)	1	300	0.00	60,000.00	5000.0
	2	0	0.00	0.00	0.0
	3				0.0
	4				0.0
	5				

	Position (ft)	Load (kip)
Axle Spacing:	14 ft	
Front Axle (dist):	14.0	-8.00
Rear Axle (dist):	0.0	-32.00



By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

BEAM REACTIONS

Position (X)	nforced Disp (dy)	orced Rot (theta)	Vert Reac (Ry)	loment Ract (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	100.0	0.0	Support reaction
600.00	0.00	0.00	-100.0	0.0	Support reaction

Vert Reac
(kip)
0.100
-0.100

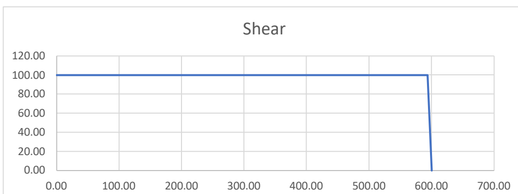
BEAM REACTIONS EQUILIBRUM CHECK

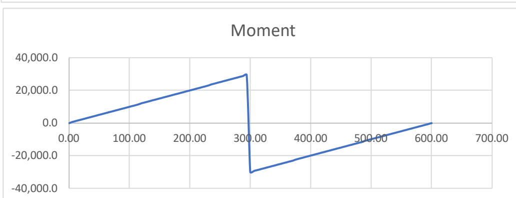
Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	0.0	0.0	0.0	OK
Sum M @ x=0	F-L	6.0000E+04	-6.0000E+04	0.0	OK

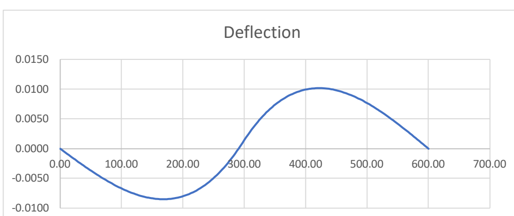
BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	0.00	100.0	0.0	0.0000
Max -V	600.00	0.0	0.0	0.0000
Max +M	294.00	100.0	29,400.0	0.0005
Max -M	300.00	100.0	-30,000.0	0.0014
Max Defl	420.00	100.0	-18,000.0	0.0102

Max/Min Value	
(---)	
0.10	kip
0.00	kip
2.45	kip-ft
-2.50	kip-ft
0.010	in







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Rev.--- By:	---	Date	---
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Macaulay Test

NON-PRINTING GRAPH AND REFERENCE DATA...DO NOT DELETE...

Beam Line

X	Y
0	0
600.00	0
0	0
0	0
0	0
0.00	0

Supports

X	Y
0.0	0
600.0	0
0.0	0
0.0	0
0.0	0

Distr Load	**1**		**2**		**3**		**4**		**5**	
No.	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**
	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0
	0.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00
	600.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00
	600.0	0	0.0	0	0.0	0	0.0	0	0.0	0

##########

Point Load No.	**Point Load 1**		**2**		**3**		**4**		**5**	
	X	**Load (Y)**	**X**	**Load (Y)**	**X**	**Load (Y)**	**X**	**Load (Y)**	**X**	**Load (Y)**
	300	0.00	0	0.00	0	0.00	0	0.00	0	0.00
	300	0.00	0	0.00	0	0.00	0	0.00	0	0.00

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Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

	X (desired)	Position (X)	Area (A)	Shear (V)	Moment (M)	Stiff. (EI)	Curv. (M/EI)	RAW Slope	RAW Defl.	Slope (theta)	Defl. (delta)
	in	L	L	F	F-L	FL^2	1/L	---	L	rad	L
0.00	0.00	0.00	17.90	100.00	0.0	1.8560E+10	0.0000E+00	0.0000E+00	0.0000	-7.59E-05	0.0000
0.01	6.00	6.00	17.90	100.00	600.0	1.8560E+10	3.2328E-08	9.6983E-08	0.0000	-7.58E-05	-0.0005
0.02	12.00	12.00	17.90	100.00	1,200.0	1.8560E+10	6.4655E-08	3.8793E-07	0.0000	-7.55E-05	-0.0009
0.03	18.00	18.00	17.90	100.00	1,800.0	1.8560E+10	9.6983E-08	8.7284E-07	0.0000	-7.51E-05	-0.0014
0.04	24.00	24.00	17.90	100.00	2,400.0	1.8560E+10	1.2931E-07	1.5517E-06	0.0000	-7.44E-05	-0.0018
0.05	30.00	30.00	17.90	100.00	3,000.0	1.8560E+10	1.6164E-07	2.4246E-06	0.0000	-7.35E-05	-0.0023
0.06	36.00	36.00	17.90	100.00	3,600.0	1.8560E+10	1.9397E-07	3.4914E-06	0.0000	-7.24E-05	-0.0027
0.07	42.00	42.00	17.90	100.00	4,200.0	1.8560E+10	2.2629E-07	4.7522E-06	0.0001	-7.12E-05	-0.0031
0.08	48.00	48.00	17.90	100.00	4,800.0	1.8560E+10	2.5862E-07	6.2069E-06	0.0001	-6.97E-05	-0.0035
0.09	54.00	54.00	17.90	100.00	5,400.0	1.8560E+10	2.9095E-07	7.8556E-06	0.0001	-6.81E-05	-0.0040
0.10	60.00	60.00	17.90	100.00	6,000.0	1.8560E+10	3.2328E-07	9.6983E-06	0.0002	-6.62E-05	-0.0044
0.11	66.00	66.00	17.90	100.00	6,600.0	1.8560E+10	3.5560E-07	1.1735E-05	0.0003	-6.42E-05	-0.0048
0.12	72.00	72.00	17.90	100.00	7,200.0	1.8560E+10	3.8793E-07	1.3966E-05	0.0003	-6.20E-05	-0.0051
0.13	78.00	78.00	17.90	100.00	7,800.0	1.8560E+10	4.2026E-07	1.6390E-05	0.0004	-5.95E-05	-0.0055
0.14	84.00	84.00	17.90	100.00	8,400.0	1.8560E+10	4.5259E-07	1.9009E-05	0.0005	-5.69E-05	-0.0058
0.15	90.00	90.00	17.90	100.00	9,000.0	1.8560E+10	4.8491E-07	2.1821E-05	0.0007	-5.41E-05	-0.0062
0.16	96.00	96.00	17.90	100.00	9,600.0	1.8560E+10	5.1724E-07	2.4828E-05	0.0008	-5.11E-05	-0.0065
0.17	102.00	102.00	17.90	100.00	10,200.0	1.8560E+10	5.4957E-07	2.8028E-05	0.0010	-4.79E-05	-0.0068
0.18	108.00	108.00	17.90	100.00	10,800.0	1.8560E+10	5.8190E-07	3.1422E-05	0.0011	-4.45E-05	-0.0071
0.19	114.00	114.00	17.90	100.00	11,400.0	1.8560E+10	6.1422E-07	3.5011E-05	0.0013	-4.09E-05	-0.0073
0.20	120.00	120.00	17.90	100.00	12,000.0	1.8560E+10	6.4655E-07	3.8793E-05	0.0016	-3.71E-05	-0.0076
0.21	126.00	126.00	17.90	100.00	12,600.0	1.8560E+10	6.7888E-07	4.2769E-05	0.0018	-3.32E-05	-0.0078
0.22	132.00	132.00	17.90	100.00	13,200.0	1.8560E+10	7.1121E-07	4.6940E-05	0.0021	-2.90E-05	-0.0080
0.23	138.00	138.00	17.90	100.00	13,800.0	1.8560E+10	7.4353E-07	5.1304E-05	0.0024	-2.46E-05	-0.0081
0.24	144.00	144.00	17.90	100.00	14,400.0	1.8560E+10	7.7586E-07	5.5862E-05	0.0027	-2.01E-05	-0.0083
0.25	150.00	150.00	17.90	100.00	15,000.0	1.8560E+10	8.0819E-07	6.0614E-05	0.0030	-1.53E-05	-0.0084
0.26	156.00	156.00	17.90	100.00	15,600.0	1.8560E+10	8.4052E-07	6.5560E-05	0.0034	-1.04E-05	-0.0084
0.27	162.00	162.00	17.90	100.00	16,200.0	1.8560E+10	8.7284E-07	7.0700E-05	0.0038	-5.24E-06	-0.0085
0.28	168.00	168.00	17.90	100.00	16,800.0	1.8560E+10	9.0517E-07	7.6034E-05	0.0043	9.70E-08	-0.0085
0.29	174.00	174.00	17.90	100.00	17,400.0	1.8560E+10	9.3750E-07	8.1563E-05	0.0047	5.62E-06	-0.0085
0.30	180.00	180.00	17.90	100.00	18,000.0	1.8560E+10	9.6983E-07	8.7284E-05	0.0052	1.13E-05	-0.0084
0.31	186.00	186.00	17.90	100.00	18,600.0	1.8560E+10	1.0022E-06	9.3200E-05	0.0058	1.73E-05	-0.0083
0.32	192.00	192.00	17.90	100.00	19,200.0	1.8560E+10	1.0345E-06	9.9310E-05	0.0064	2.34E-05	-0.0082
0.33	198.00	198.00	17.90	100.00	19,800.0	1.8560E+10	1.0668E-06	1.0561E-04	0.0070	2.97E-05	-0.0081
0.34	204.00	204.00	17.90	100.00	20,400.0	1.8560E+10	1.0991E-06	1.1211E-04	0.0076	3.62E-05	-0.0079
0.35	210.00	210.00	17.90	100.00	21,000.0	1.8560E+10	1.1315E-06	1.1880E-04	0.0083	4.29E-05	-0.0076
0.36	216.00	216.00	17.90	100.00	21,600.0	1.8560E+10	1.1638E-06	1.2569E-04	0.0091	4.98E-05	-0.0073
0.37	222.00	222.00	17.90	100.00	22,200.0	1.8560E+10	1.1961E-06	1.3277E-04	0.0098	5.68E-05	-0.0070
0.38	228.00	228.00	17.90	100.00	22,800.0	1.8560E+10	1.2284E-06	1.4004E-04	0.0106	6.41E-05	-0.0067
0.39	234.00	234.00	17.90	100.00	23,400.0	1.8560E+10	1.2608E-06	1.4751E-04	0.0115	7.16E-05	-0.0063
0.40	240.00	240.00	17.90	100.00	24,000.0	1.8560E+10	1.2931E-06	1.5517E-04	0.0124	7.92E-05	-0.0058
0.41	246.00	246.00	17.90	100.00	24,600.0	1.8560E+10	1.3254E-06	1.6303E-04	0.0134	8.71E-05	-0.0053
0.42	252.00	252.00	17.90	100.00	25,200.0	1.8560E+10	1.3578E-06	1.7108E-04	0.0144	9.51E-05	-0.0048
0.43	258.00	258.00	17.90	100.00	25,800.0	1.8560E+10	1.3901E-06	1.7932E-04	0.0154	1.03E-04	-0.0042
0.44	264.00	264.00	17.90	100.00	26,400.0	1.8560E+10	1.4224E-06	1.8776E-04	0.0165	1.12E-04	-0.0035
0.45	270.00	270.00	17.90	100.00	27,000.0	1.8560E+10	1.4547E-06	1.9639E-04	0.0177	1.20E-04	-0.0028
0.46	276.00	276.00	17.90	100.00	27,600.0	1.8560E+10	1.4871E-06	2.0522E-04	0.0189	1.29E-04	-0.0021
0.47	282.00	282.00	17.90	100.00	28,200.0	1.8560E+10	1.5194E-06	2.1423E-04	0.0201	1.38E-04	-0.0013
0.48	288.00	288.00	17.90	100.00	28,800.0	1.8560E+10	1.5517E-06	2.2345E-04	0.0215	1.48E-04	-0.0004
0.49	294.00	294.00	17.90	100.00	29,400.0	1.8560E+10	1.5841E-06	2.3286E-04	0.0228	1.57E-04	0.0005
0.50	300.00	300.00	17.90	100.00	-30,000.0	1.8560E+10	-1.6164E-06	2.3276E-04	0.0242	1.57E-04	0.0014
0.51	306.00	306.00	17.90	100.00	-29,400.0	1.8560E+10	-1.5841E-06	2.2316E-04	0.0256	1.47E-04	0.0024
0.52	312.00	312.00	17.90	100.00	-28,800.0	1.8560E+10	-1.5517E-06	2.1375E-04	0.0269	1.38E-04	0.0032
0.53	318.00	318.00	17.90	100.00	-28,200.0	1.8560E+10	-1.5194E-06	2.0454E-04	0.0282	1.29E-04	0.0040
0.54	324.00	324.00	17.90	100.00	-27,600.0	1.8560E+10	-1.4871E-06	1.9552E-04	0.0294	1.20E-04	0.0048
0.55	330.00	330.00	17.90	100.00	-27,000.0	1.8560E+10	-1.4547E-06	1.8669E-04	0.0305	1.11E-04	0.0054
0.56	336.00	336.00	17.90	100.00	-26,400.0	1.8560E+10	-1.4224E-06	1.7806E-04	0.0316	1.02E-04	0.0061
0.57	342.00	342.00	17.90	100.00	-25,800.0	1.8560E+10	-1.3901E-06	1.6962E-04	0.0326	9.37E-05	0.0067
0.58	348.00	348.00	17.90	100.00	-25,200.0	1.8560E+10	-1.3578E-06	1.6138E-04	0.0336	8.54E-05	0.0072
0.59	354.00	354.00	17.90	100.00	-24,600.0	1.8560E+10	-1.3254E-06	1.5333E-04	0.0346	7.74E-05	0.0077
0.60	360.00	360.00	17.90	100.00	-24,000.0	1.8560E+10	-1.2931E-06	1.4547E-04	0.0355	6.95E-05	0.0081
0.61	366.00	366.00	17.90	100.00	-23,400.0	1.8560E+10	-1.2608E-06	1.3781E-04	0.0363	6.19E-05	0.0085
0.62	372.00	372.00	17.90	100.00	-22,800.0	1.8560E+10	-1.2284E-06	1.3034E-04	0.0371	5.44E-05	0.0089
0.63	378.00	378.00	17.90	100.00	-22,200.0	1.8560E+10	-1.1961E-06	1.2307E-04	0.0379	4.71E-05	0.0092
0.64	384.00	384.00	17.90	100.00	-21,600.0	1.8560E+10	-1.1638E-06	1.1599E-04	0.0386	4.01E-05	0.0094
0.65	390.00	390.00	17.90	100.00	-21,000.0	1.8560E+10	-1.1315E-06	1.0911E-04	0.0393	3.32E-05	0.0097
0.66	396.00	396.00	17.90	100.00	-20,400.0	1.8560E+10	-1.0991E-06	1.0241E-04	0.0399	2.65E-05	0.0098
0.67	402.00	402.00	17.90	100.00	-19,800.0	1.8560E+10	-1.0668E-06	9.5916E-05	0.0405	2.00E-05	0.0100
0.68	408.00	408.00	17.90	100.00	-19,200.0	1.8560E+10	-1.0345E-06	8.9612E-05	0.0411	1.37E-05	0.0101
0.69	414.00	414.00	17.90	100.00	-18,600.0	1.8560E+10	-1.0022E-06	8.3502E-05	0.0416	7.56E-06	0.0101
0.70	420.00	420.00	17.90	100.00	-18,000.0	1.8560E+10	-9.6983E-07	7.7586E-05	0.0421	1.65E-06	0.0102
0.71	426.00	426.00	17.90	100.00	-17,400.0	1.8560E+10	-9.3750E-07	7.1864E-05	0.0425	-4.07E-06	0.0102
0.72	432.00	432.00	17.90	100.00	-16,800.0	1.8560E+10	-9.0517E-07	6.6336E-05	0.0429	-9.60E-06	0.0101
0.73	438.00	438.00	17.90	100.00	-16,200.0	1.8560E+10	-8.7284E-07	6.1002E-05	0.0433	-1.49E-05	0.0101
0.74	444.00	444.00	17.90	100.00	-15,600.0	1.8560E+10	-8.4052E-07	5.5862E-05	0.0437	-2.01E-05	0.0099

By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
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Macaulay Test

0.75	450.00	450.00	17.90	100.00	-15,000.0	1.8560E+10	-8.0819E-07	5.0916E-05	0.0440	-2.50E-05	0.0098
0.76	456.00	456.00	17.90	100.00	-14,400.0	1.8560E+10	-7.7586E-07	4.6164E-05	0.0443	-2.98E-05	0.0096
0.77	462.00	462.00	17.90	100.00	-13,800.0	1.8560E+10	-7.4353E-07	4.1606E-05	0.0445	-3.43E-05	0.0095
0.78	468.00	468.00	17.90	100.00	-13,200.0	1.8560E+10	-7.1121E-07	3.7241E-05	0.0448	-3.87E-05	0.0092
0.79	474.00	474.00	17.90	100.00	-12,600.0	1.8560E+10	-6.7888E-07	3.3071E-05	0.0450	-4.29E-05	0.0090
0.80	480.00	480.00	17.90	100.00	-12,000.0	1.8560E+10	-6.4655E-07	2.9095E-05	0.0452	-4.68E-05	0.0087
0.81	486.00	486.00	17.90	100.00	-11,400.0	1.8560E+10	-6.1422E-07	2.5313E-05	0.0453	-5.06E-05	0.0084
0.82	492.00	492.00	17.90	100.00	-10,800.0	1.8560E+10	-5.8190E-07	2.1724E-05	0.0455	-5.42E-05	0.0081
0.83	498.00	498.00	17.90	100.00	-10,200.0	1.8560E+10	-5.4957E-07	1.8330E-05	0.0456	-5.76E-05	0.0078
0.84	504.00	504.00	17.90	100.00	-9,600.0	1.8560E+10	-5.1724E-07	1.5129E-05	0.0457	-6.08E-05	0.0074
0.85	510.00	510.00	17.90	100.00	-9,000.0	1.8560E+10	-4.8491E-07	1.2123E-05	0.0458	-6.38E-05	0.0071
0.86	516.00	516.00	17.90	100.00	-8,400.0	1.8560E+10	-4.5259E-07	9.3103E-06	0.0458	-6.66E-05	0.0067
0.87	522.00	522.00	17.90	100.00	-7,800.0	1.8560E+10	-4.2026E-07	6.6918E-06	0.0459	-6.92E-05	0.0063
0.88	528.00	528.00	17.90	100.00	-7,200.0	1.8560E+10	-3.8793E-07	4.2672E-06	0.0459	-7.17E-05	0.0058
0.89	534.00	534.00	17.90	100.00	-6,600.0	1.8560E+10	-3.5560E-07	2.0366E-06	0.0459	-7.39E-05	0.0054
0.90	540.00	540.00	17.90	100.00	-6,000.0	1.8560E+10	-3.2328E-07	-2.3823E-21	0.0459	-7.59E-05	0.0049
0.91	546.00	546.00	17.90	100.00	-5,400.0	1.8560E+10	-2.9095E-07	-1.8427E-06	0.0459	-7.78E-05	0.0045
0.92	552.00	552.00	17.90	100.00	-4,800.0	1.8560E+10	-2.5862E-07	-3.4914E-06	0.0459	-7.94E-05	0.0040
0.93	558.00	558.00	17.90	100.00	-4,200.0	1.8560E+10	-2.2629E-07	-4.9461E-06	0.0459	-8.09E-05	0.0035
0.94	564.00	564.00	17.90	100.00	-3,600.0	1.8560E+10	-1.9397E-07	-6.2069E-06	0.0459	-8.21E-05	0.0030
0.95	570.00	570.00	17.90	100.00	-3,000.0	1.8560E+10	-1.6164E-07	-7.2737E-06	0.0458	-8.32E-05	0.0025
0.96	576.00	576.00	17.90	100.00	-2,400.0	1.8560E+10	-1.2931E-07	-8.1466E-06	0.0458	-8.41E-05	0.0020
0.97	582.00	582.00	17.90	100.00	-1,800.0	1.8560E+10	-9.6983E-08	-8.8254E-06	0.0457	-8.48E-05	0.0015
0.98	588.00	588.00	17.90	100.00	-1,200.0	1.8560E+10	-6.4655E-08	-9.3103E-06	0.0457	-8.52E-05	0.0010
0.99	594.00	594.00	17.90	100.00	-600.0	1.8560E+10	-3.2328E-08	-9.6013E-06	0.0456	-8.55E-05	0.0005
1.00	600.00	600.00	17.90	0.00	0.0	1.8560E+10	0.0000E+00	-9.6983E-06	0.0456	-8.56E-05	0.0000

By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

Macaulay Beam Analysis - BetaTesting

13 - Steel - Single Span -Moment

Bridge, L =		in		E =	29,000,000	lb / in^2	Length Units	in
Beam Sect =	W14x61	W14x90		G =	1.740E+06	lb / in^2 = 0.06*29,000,000	Force Units	lb
A =	17.90	26.50	in^2					
I =	640.00	999.00	in^4					

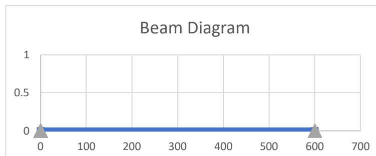
Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in^2	in^4	lb / in^2	lb / in^2
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

	Supp. No.	X	Imposed Movement * dY (vert)	** dr (rot)
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	0.0	0.000
	3			
	4			
	5			

* (settlement) ** (Not Working)



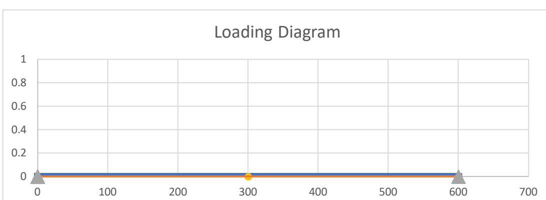
Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)	Load
		in	in	lb / in	lb / in	lb / ft
DO NOT DELETE ROWS (GRAPHED)	1	0.0	600.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0	0.0
	5					

Point Loads - UNFACTORED

	Load No.	X	Py	My	Load
		in	lb	lb-in	lb-ft
DO NOT DELETE ROWS (GRAPHED)	1	300	0.00	60,000.00	5000.0
	2	0	0.00	0.00	0.0
	3				0.0
	4				0.0
	5				

Axle Spacing:	14 ft	
	Position (ft)	Load (kip)
Front Axle (dist):	14.0	-8.00
Rear Axle (dist):	0.0	-32.00



By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

BEAM REACTIONS

Position (X)	nforced Disp (dy)	orced Rot (theta)	Vert Reac (Ry)	loment Ract (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	100.0	0.0	Support reaction
600.00	0.00	0.00	-100.0	0.0	Support reaction

Vert Reac
(kip)
0.100
-0.100

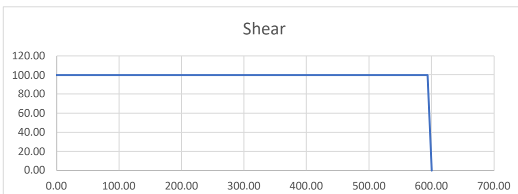
BEAM REACTIONS EQUILIBRUM CHECK

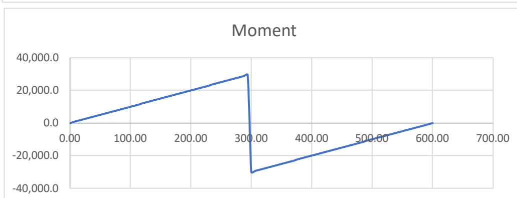
Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	0.0	0.0	0.0	OK
Sum M @ x=0	F-L	6.0000E+04	-6.0000E+04	0.0	OK

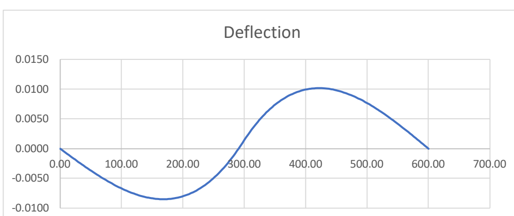
BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	0.00	100.0	0.0	0.0000
Max -V	600.00	0.0	0.0	0.0000
Max +M	294.00	100.0	29,400.0	0.0005
Max -M	300.00	100.0	-30,000.0	0.0014
Max Defl	420.00	100.0	-18,000.0	0.0102

Max/Min Value	
(---)	
0.10	kip
0.00	kip
2.45	kip-ft
-2.50	kip-ft
0.010	in







By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

NON-PRINTING GRAPH AND REFERENCE DATA...DO NOT DELETE...

Beam Line

X	Y
0	0
600.00	0
0	0
0	0
0	0
0.00	0

Supports

X	Y
0.0	0
600.0	0
0.0	0
0.0	0
0.0	0

Distr Load	**1**		**2**		**3**		**4**		**5**	
No.	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**	**(X)**	**Load (Y)**
	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0
	0.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00
	600.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00	0.0	0.00
	600.0	0	0.0	0	0.0	0	0.0	0	0.0	0

##########

Point Load No.	**Point Load 1**		**2**		**3**		**4**		**5**	
	X	**Load (Y)**	**X**	**Load (Y)**	**X**	**Load (Y)**	**X**	**Load (Y)**	**X**	**Load (Y)**
	300	0.00	0	0.00	0	0.00	0	0.00	0	0.00
	300	0.00	0	0.00	0	0.00	0	0.00	0	0.00

By: AKH   Date May 2026
Rev.--- By: ---   Date ---
Chk: Date

Macaulay Test

	X (desired)	Position (X)	Area (A)	Shear (V)	Moment (M)	Stiff. (EI)	Curv. (M/EI)	RAW Slope	RAW Defl.	Slope (theta)	Defl. (delta)
	in	L	L	F	F-L	FL^2	1/L	---	L	rad	L
0.00	0.00	0.00	17.90	100.00	0.0	1.8560E+10	0.0000E+00	0.0000E+00	0.0000	-7.59E-05	0.0000
0.01	6.00	6.00	17.90	100.00	600.0	1.8560E+10	3.2328E-08	9.6983E-08	0.0000	-7.58E-05	-0.0005
0.02	12.00	12.00	17.90	100.00	1,200.0	1.8560E+10	6.4655E-08	3.8793E-07	0.0000	-7.55E-05	-0.0009
0.03	18.00	18.00	17.90	100.00	1,800.0	1.8560E+10	9.6983E-08	8.7284E-07	0.0000	-7.51E-05	-0.0014
0.04	24.00	24.00	17.90	100.00	2,400.0	1.8560E+10	1.2931E-07	1.5517E-06	0.0000	-7.44E-05	-0.0018
0.05	30.00	30.00	17.90	100.00	3,000.0	1.8560E+10	1.6164E-07	2.4246E-06	0.0000	-7.35E-05	-0.0023
0.06	36.00	36.00	17.90	100.00	3,600.0	1.8560E+10	1.9397E-07	3.4914E-06	0.0000	-7.24E-05	-0.0027
0.07	42.00	42.00	17.90	100.00	4,200.0	1.8560E+10	2.2629E-07	4.7522E-06	0.0001	-7.12E-05	-0.0031
0.08	48.00	48.00	17.90	100.00	4,800.0	1.8560E+10	2.5862E-07	6.2069E-06	0.0001	-6.97E-05	-0.0035
0.09	54.00	54.00	17.90	100.00	5,400.0	1.8560E+10	2.9095E-07	7.8556E-06	0.0001	-6.81E-05	-0.0040
0.10	60.00	60.00	17.90	100.00	6,000.0	1.8560E+10	3.2328E-07	9.6983E-06	0.0002	-6.62E-05	-0.0044
0.11	66.00	66.00	17.90	100.00	6,600.0	1.8560E+10	3.5560E-07	1.1735E-05	0.0003	-6.42E-05	-0.0048
0.12	72.00	72.00	17.90	100.00	7,200.0	1.8560E+10	3.8793E-07	1.3966E-05	0.0003	-6.20E-05	-0.0051
0.13	78.00	78.00	17.90	100.00	7,800.0	1.8560E+10	4.2026E-07	1.6390E-05	0.0004	-5.95E-05	-0.0055
0.14	84.00	84.00	17.90	100.00	8,400.0	1.8560E+10	4.5259E-07	1.9009E-05	0.0005	-5.69E-05	-0.0058
0.15	90.00	90.00	17.90	100.00	9,000.0	1.8560E+10	4.8491E-07	2.1821E-05	0.0007	-5.41E-05	-0.0062
0.16	96.00	96.00	17.90	100.00	9,600.0	1.8560E+10	5.1724E-07	2.4828E-05	0.0008	-5.11E-05	-0.0065
0.17	102.00	102.00	17.90	100.00	10,200.0	1.8560E+10	5.4957E-07	2.8028E-05	0.0010	-4.79E-05	-0.0068
0.18	108.00	108.00	17.90	100.00	10,800.0	1.8560E+10	5.8190E-07	3.1422E-05	0.0011	-4.45E-05	-0.0071
0.19	114.00	114.00	17.90	100.00	11,400.0	1.8560E+10	6.1422E-07	3.5011E-05	0.0013	-4.09E-05	-0.0073
0.20	120.00	120.00	17.90	100.00	12,000.0	1.8560E+10	6.4655E-07	3.8793E-05	0.0016	-3.71E-05	-0.0076
0.21	126.00	126.00	17.90	100.00	12,600.0	1.8560E+10	6.7888E-07	4.2769E-05	0.0018	-3.32E-05	-0.0078
0.22	132.00	132.00	17.90	100.00	13,200.0	1.8560E+10	7.1121E-07	4.6940E-05	0.0021	-2.90E-05	-0.0080
0.23	138.00	138.00	17.90	100.00	13,800.0	1.8560E+10	7.4353E-07	5.1304E-05	0.0024	-2.46E-05	-0.0081
0.24	144.00	144.00	17.90	100.00	14,400.0	1.8560E+10	7.7586E-07	5.5862E-05	0.0027	-2.01E-05	-0.0083
0.25	150.00	150.00	17.90	100.00	15,000.0	1.8560E+10	8.0819E-07	6.0614E-05	0.0030	-1.53E-05	-0.0084
0.26	156.00	156.00	17.90	100.00	15,600.0	1.8560E+10	8.4052E-07	6.5560E-05	0.0034	-1.04E-05	-0.0084
0.27	162.00	162.00	17.90	100.00	16,200.0	1.8560E+10	8.7284E-07	7.0700E-05	0.0038	-5.24E-06	-0.0085
0.28	168.00	168.00	17.90	100.00	16,800.0	1.8560E+10	9.0517E-07	7.6034E-05	0.0043	9.70E-08	-0.0085
0.29	174.00	174.00	17.90	100.00	17,400.0	1.8560E+10	9.3750E-07	8.1563E-05	0.0047	5.62E-06	-0.0085
0.30	180.00	180.00	17.90	100.00	18,000.0	1.8560E+10	9.6983E-07	8.7284E-05	0.0052	1.13E-05	-0.0084
0.31	186.00	186.00	17.90	100.00	18,600.0	1.8560E+10	1.0022E-06	9.3200E-05	0.0058	1.73E-05	-0.0083
0.32	192.00	192.00	17.90	100.00	19,200.0	1.8560E+10	1.0345E-06	9.9310E-05	0.0064	2.34E-05	-0.0082
0.33	198.00	198.00	17.90	100.00	19,800.0	1.8560E+10	1.0668E-06	1.0561E-04	0.0070	2.97E-05	-0.0081
0.34	204.00	204.00	17.90	100.00	20,400.0	1.8560E+10	1.0991E-06	1.1211E-04	0.0076	3.62E-05	-0.0079
0.35	210.00	210.00	17.90	100.00	21,000.0	1.8560E+10	1.1315E-06	1.1880E-04	0.0083	4.29E-05	-0.0076
0.36	216.00	216.00	17.90	100.00	21,600.0	1.8560E+10	1.1638E-06	1.2569E-04	0.0091	4.98E-05	-0.0073
0.37	222.00	222.00	17.90	100.00	22,200.0	1.8560E+10	1.1961E-06	1.3277E-04	0.0098	5.68E-05	-0.0070
0.38	228.00	228.00	17.90	100.00	22,800.0	1.8560E+10	1.2284E-06	1.4004E-04	0.0106	6.41E-05	-0.0067
0.39	234.00	234.00	17.90	100.00	23,400.0	1.8560E+10	1.2608E-06	1.4751E-04	0.0115	7.16E-05	-0.0063
0.40	240.00	240.00	17.90	100.00	24,000.0	1.8560E+10	1.2931E-06	1.5517E-04	0.0124	7.92E-05	-0.0058
0.41	246.00	246.00	17.90	100.00	24,600.0	1.8560E+10	1.3254E-06	1.6303E-04	0.0134	8.71E-05	-0.0053
0.42	252.00	252.00	17.90	100.00	25,200.0	1.8560E+10	1.3578E-06	1.7108E-04	0.0144	9.51E-05	-0.0048
0.43	258.00	258.00	17.90	100.00	25,800.0	1.8560E+10	1.3901E-06	1.7932E-04	0.0154	1.03E-04	-0.0042
0.44	264.00	264.00	17.90	100.00	26,400.0	1.8560E+10	1.4224E-06	1.8776E-04	0.0165	1.12E-04	-0.0035
0.45	270.00	270.00	17.90	100.00	27,000.0	1.8560E+10	1.4547E-06	1.9639E-04	0.0177	1.20E-04	-0.0028
0.46	276.00	276.00	17.90	100.00	27,600.0	1.8560E+10	1.4871E-06	2.0522E-04	0.0189	1.29E-04	-0.0021
0.47	282.00	282.00	17.90	100.00	28,200.0	1.8560E+10	1.5194E-06	2.1423E-04	0.0201	1.38E-04	-0.0013
0.48	288.00	288.00	17.90	100.00	28,800.0	1.8560E+10	1.5517E-06	2.2345E-04	0.0215	1.48E-04	-0.0004
0.49	294.00	294.00	17.90	100.00	29,400.0	1.8560E+10	1.5841E-06	2.3286E-04	0.0228	1.57E-04	0.0005
0.50	300.00	300.00	17.90	100.00	-30,000.0	1.8560E+10	-1.6164E-06	2.3276E-04	0.0242	1.57E-04	0.0014
0.51	306.00	306.00	17.90	100.00	-29,400.0	1.8560E+10	-1.5841E-06	2.2316E-04	0.0256	1.47E-04	0.0024
0.52	312.00	312.00	17.90	100.00	-28,800.0	1.8560E+10	-1.5517E-06	2.1375E-04	0.0269	1.38E-04	0.0032
0.53	318.00	318.00	17.90	100.00	-28,200.0	1.8560E+10	-1.5194E-06	2.0454E-04	0.0282	1.29E-04	0.0040
0.54	324.00	324.00	17.90	100.00	-27,600.0	1.8560E+10	-1.4871E-06	1.9552E-04	0.0294	1.20E-04	0.0048
0.55	330.00	330.00	17.90	100.00	-27,000.0	1.8560E+10	-1.4547E-06	1.8669E-04	0.0305	1.11E-04	0.0054
0.56	336.00	336.00	17.90	100.00	-26,400.0	1.8560E+10	-1.4224E-06	1.7806E-04	0.0316	1.02E-04	0.0061
0.57	342.00	342.00	17.90	100.00	-25,800.0	1.8560E+10	-1.3901E-06	1.6962E-04	0.0326	9.37E-05	0.0067
0.58	348.00	348.00	17.90	100.00	-25,200.0	1.8560E+10	-1.3578E-06	1.6138E-04	0.0336	8.54E-05	0.0072
0.59	354.00	354.00	17.90	100.00	-24,600.0	1.8560E+10	-1.3254E-06	1.5333E-04	0.0346	7.74E-05	0.0077
0.60	360.00	360.00	17.90	100.00	-24,000.0	1.8560E+10	-1.2931E-06	1.4547E-04	0.0355	6.95E-05	0.0081
0.61	366.00	366.00	17.90	100.00	-23,400.0	1.8560E+10	-1.2608E-06	1.3781E-04	0.0363	6.19E-05	0.0085
0.62	372.00	372.00	17.90	100.00	-22,800.0	1.8560E+10	-1.2284E-06	1.3034E-04	0.0371	5.44E-05	0.0089
0.63	378.00	378.00	17.90	100.00	-22,200.0	1.8560E+10	-1.1961E-06	1.2307E-04	0.0379	4.71E-05	0.0092
0.64	384.00	384.00	17.90	100.00	-21,600.0	1.8560E+10	-1.1638E-06	1.1599E-04	0.0386	4.01E-05	0.0094
0.65	390.00	390.00	17.90	100.00	-21,000.0	1.8560E+10	-1.1315E-06	1.0911E-04	0.0393	3.32E-05	0.0097
0.66	396.00	396.00	17.90	100.00	-20,400.0	1.8560E+10	-1.0991E-06	1.0241E-04	0.0399	2.65E-05	0.0098
0.67	402.00	402.00	17.90	100.00	-19,800.0	1.8560E+10	-1.0668E-06	9.5916E-05	0.0405	2.00E-05	0.0100
0.68	408.00	408.00	17.90	100.00	-19,200.0	1.8560E+10	-1.0345E-06	8.9612E-05	0.0411	1.37E-05	0.0101
0.69	414.00	414.00	17.90	100.00	-18,600.0	1.8560E+10	-1.0022E-06	8.3502E-05	0.0416	7.56E-06	0.0101
0.70	420.00	420.00	17.90	100.00	-18,000.0	1.8560E+10	-9.6983E-07	7.7586E-05	0.0421	1.65E-06	0.0102
0.71	426.00	426.00	17.90	100.00	-17,400.0	1.8560E+10	-9.3750E-07	7.1864E-05	0.0425	-4.07E-06	0.0102
0.72	432.00	432.00	17.90	100.00	-16,800.0	1.8560E+10	-9.0517E-07	6.6336E-05	0.0429	-9.60E-06	0.0101
0.73	438.00	438.00	17.90	100.00	-16,200.0	1.8560E+10	-8.7284E-07	6.1002E-05	0.0433	-1.49E-05	0.0101
0.74	444.00	444.00	17.90	100.00	-15,600.0	1.8560E+10	-8.4052E-07	5.5862E-05	0.0437	-2.01E-05	0.0099

By:	AKH
Rev.--- By:	---
Chk:	

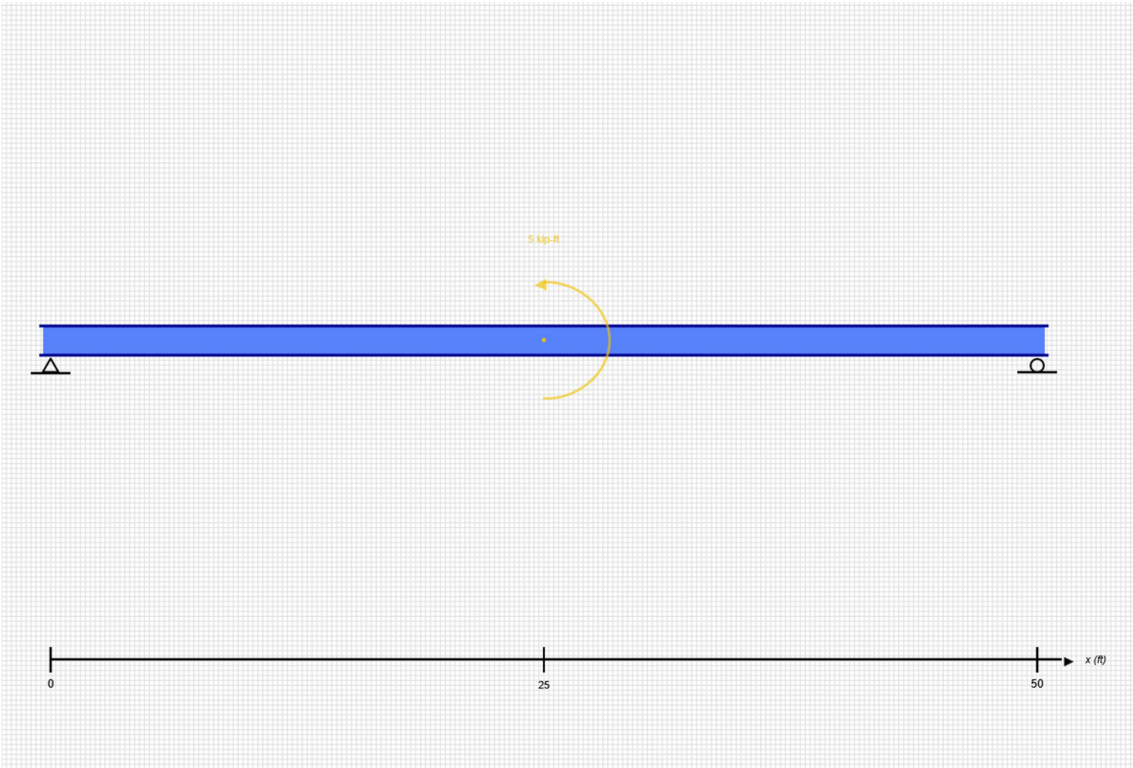
Date	May 2026
Date	---
Date	

Macaulay Test

0.75	450.00	450.00	17.90	100.00	-15,000.0	1.8560E+10	-8.0819E-07	5.0916E-05	0.0440	-2.50E-05	0.0098
0.76	456.00	456.00	17.90	100.00	-14,400.0	1.8560E+10	-7.7586E-07	4.6164E-05	0.0443	-2.98E-05	0.0096
0.77	462.00	462.00	17.90	100.00	-13,800.0	1.8560E+10	-7.4353E-07	4.1606E-05	0.0445	-3.43E-05	0.0095
0.78	468.00	468.00	17.90	100.00	-13,200.0	1.8560E+10	-7.1121E-07	3.7241E-05	0.0448	-3.87E-05	0.0092
0.79	474.00	474.00	17.90	100.00	-12,600.0	1.8560E+10	-6.7888E-07	3.3071E-05	0.0450	-4.29E-05	0.0090
0.80	480.00	480.00	17.90	100.00	-12,000.0	1.8560E+10	-6.4655E-07	2.9095E-05	0.0452	-4.68E-05	0.0087
0.81	486.00	486.00	17.90	100.00	-11,400.0	1.8560E+10	-6.1422E-07	2.5313E-05	0.0453	-5.06E-05	0.0084
0.82	492.00	492.00	17.90	100.00	-10,800.0	1.8560E+10	-5.8190E-07	2.1724E-05	0.0455	-5.42E-05	0.0081
0.83	498.00	498.00	17.90	100.00	-10,200.0	1.8560E+10	-5.4957E-07	1.8330E-05	0.0456	-5.76E-05	0.0078
0.84	504.00	504.00	17.90	100.00	-9,600.0	1.8560E+10	-5.1724E-07	1.5129E-05	0.0457	-6.08E-05	0.0074
0.85	510.00	510.00	17.90	100.00	-9,000.0	1.8560E+10	-4.8491E-07	1.2123E-05	0.0458	-6.38E-05	0.0071
0.86	516.00	516.00	17.90	100.00	-8,400.0	1.8560E+10	-4.5259E-07	9.3103E-06	0.0458	-6.66E-05	0.0067
0.87	522.00	522.00	17.90	100.00	-7,800.0	1.8560E+10	-4.2026E-07	6.6918E-06	0.0459	-6.92E-05	0.0063
0.88	528.00	528.00	17.90	100.00	-7,200.0	1.8560E+10	-3.8793E-07	4.2672E-06	0.0459	-7.17E-05	0.0058
0.89	534.00	534.00	17.90	100.00	-6,600.0	1.8560E+10	-3.5560E-07	2.0366E-06	0.0459	-7.39E-05	0.0054
0.90	540.00	540.00	17.90	100.00	-6,000.0	1.8560E+10	-3.2328E-07	-2.3823E-21	0.0459	-7.59E-05	0.0049
0.91	546.00	546.00	17.90	100.00	-5,400.0	1.8560E+10	-2.9095E-07	-1.8427E-06	0.0459	-7.78E-05	0.0045
0.92	552.00	552.00	17.90	100.00	-4,800.0	1.8560E+10	-2.5862E-07	-3.4914E-06	0.0459	-7.94E-05	0.0040
0.93	558.00	558.00	17.90	100.00	-4,200.0	1.8560E+10	-2.2629E-07	-4.9461E-06	0.0459	-8.09E-05	0.0035
0.94	564.00	564.00	17.90	100.00	-3,600.0	1.8560E+10	-1.9397E-07	-6.2069E-06	0.0459	-8.21E-05	0.0030
0.95	570.00	570.00	17.90	100.00	-3,000.0	1.8560E+10	-1.6164E-07	-7.2737E-06	0.0458	-8.32E-05	0.0025
0.96	576.00	576.00	17.90	100.00	-2,400.0	1.8560E+10	-1.2931E-07	-8.1466E-06	0.0458	-8.41E-05	0.0020
0.97	582.00	582.00	17.90	100.00	-1,800.0	1.8560E+10	-9.6983E-08	-8.8254E-06	0.0457	-8.48E-05	0.0015
0.98	588.00	588.00	17.90	100.00	-1,200.0	1.8560E+10	-6.4655E-08	-9.3103E-06	0.0457	-8.52E-05	0.0010
0.99	594.00	594.00	17.90	100.00	-600.0	1.8560E+10	-3.2328E-08	-9.6013E-06	0.0456	-8.55E-05	0.0005
1.00	600.00	600.00	17.90	0.00	0.0	1.8560E+10	0.0000E+00	-9.6983E-06	0.0456	-8.56E-05	0.0000

SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Wed Jun 10 2026 15:59:32 GMT-0400 (Eastern Daylight Time)

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

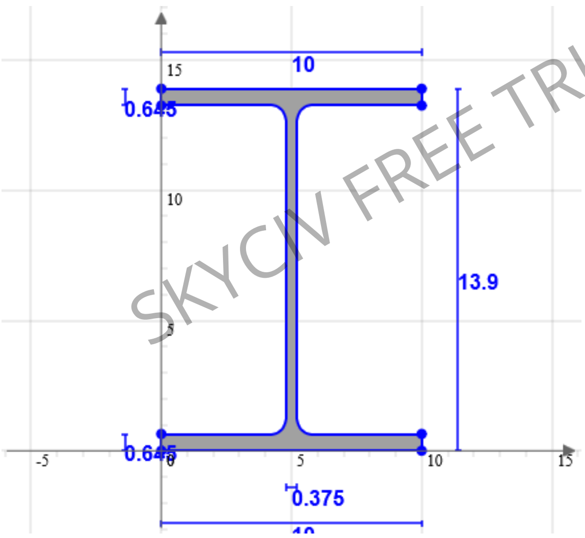
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft

Loads

Load Type	Location	Magnitude	Load Case
Moment Load	25 ft	5 kip-ft	DL

Beam Section



Geometric Properties		
A	17.9	in^2
C_z	5	in
C_y	6.95	in

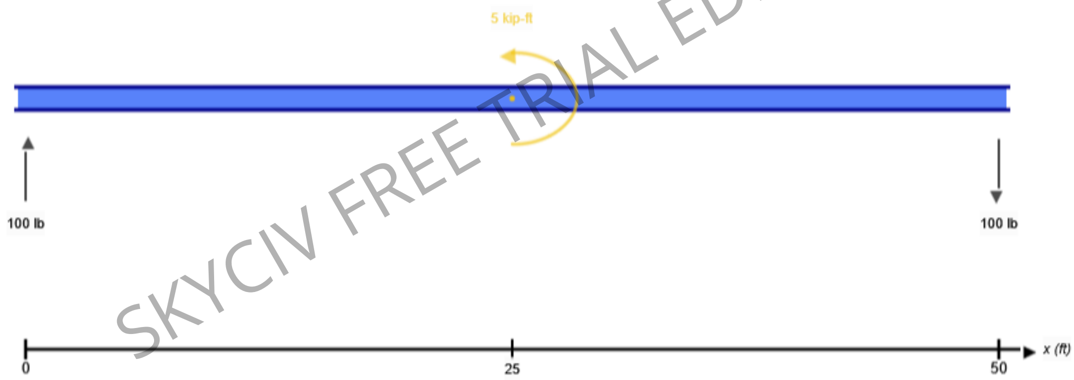
Bending Properties		
I_z	640	in^4
I_y	107	in^4

Shear Properties		
A_z	11.277	in^2
A_y	4.99	in^2

Torsion Properties		
J	2.19	in^4
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft^3)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.008	L/32291
Custom Stress Limit	PASS	39 ksi	0.008	0.326 ksi
Material Yield	PASS	36 ksi	0.009	0.326 ksi
Material Strength	PASS	58 ksi	0.006	0.326 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	100	0
50	0	-100	0

Force Extremes

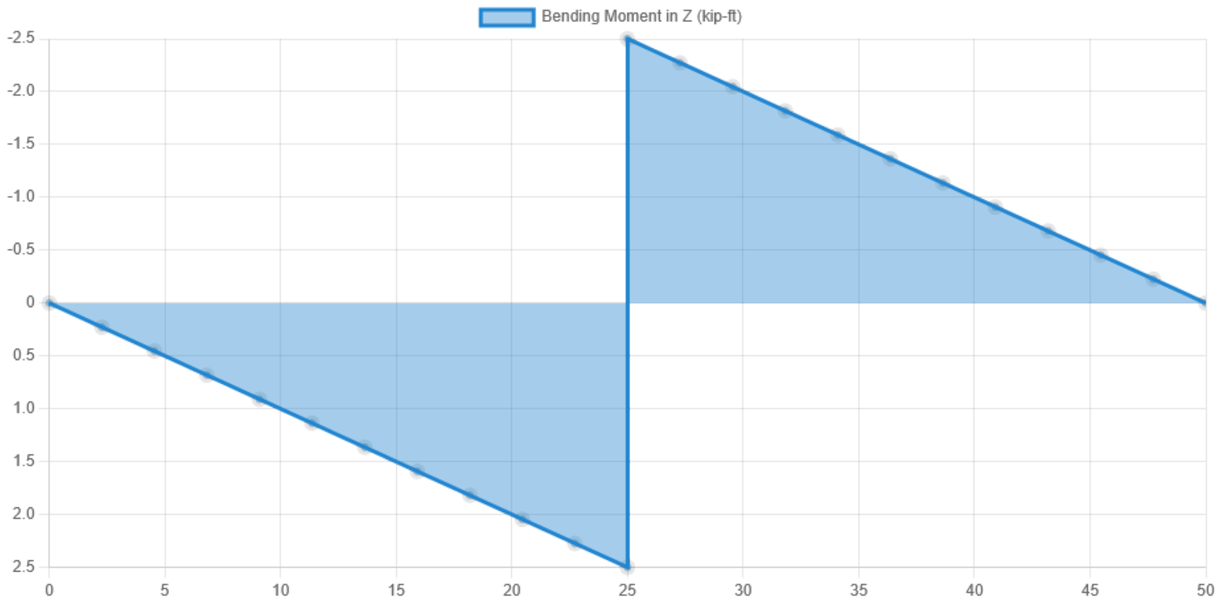
Result	Max	Min
Bending Moment	2.5 kip-ft	-2.5 kip-ft
Shear	100 lb	100 lb
Displacement Y	0.009 in	-0.009 in

Stress Extremes

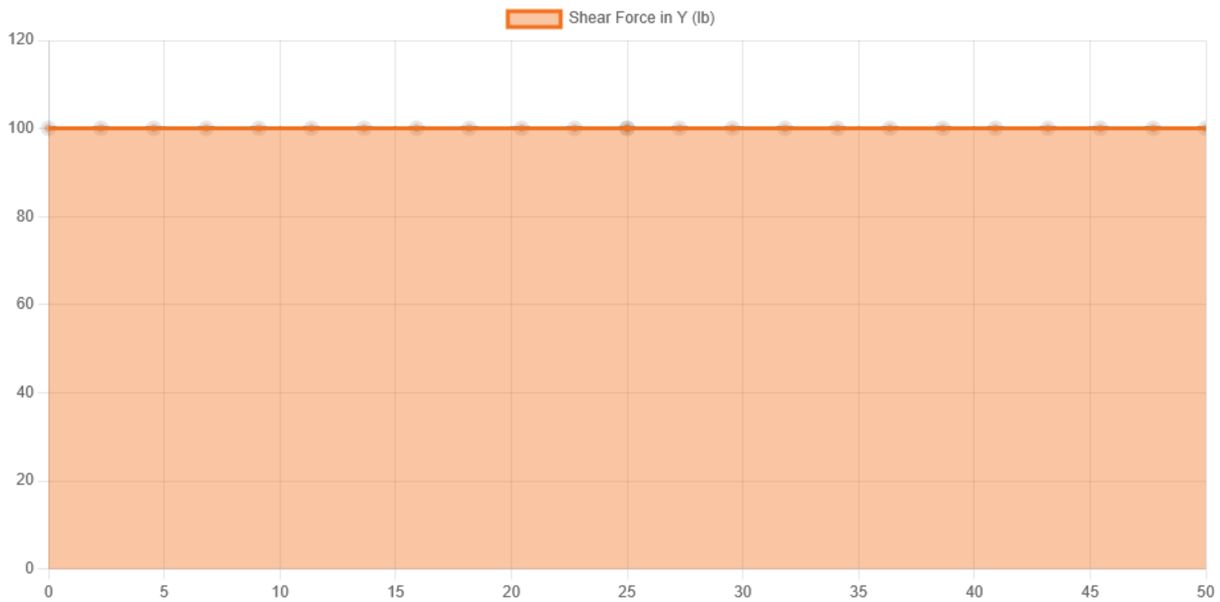
Result	Max (ksi)	Min (ksi)
Bending Stress	0.326	-0.326
Shear Stress Total	0.021	0.021
Max Combined Normal Stress	0.326	0
Min Combined Normal Stress	0	-0.326

DIAGRAMS

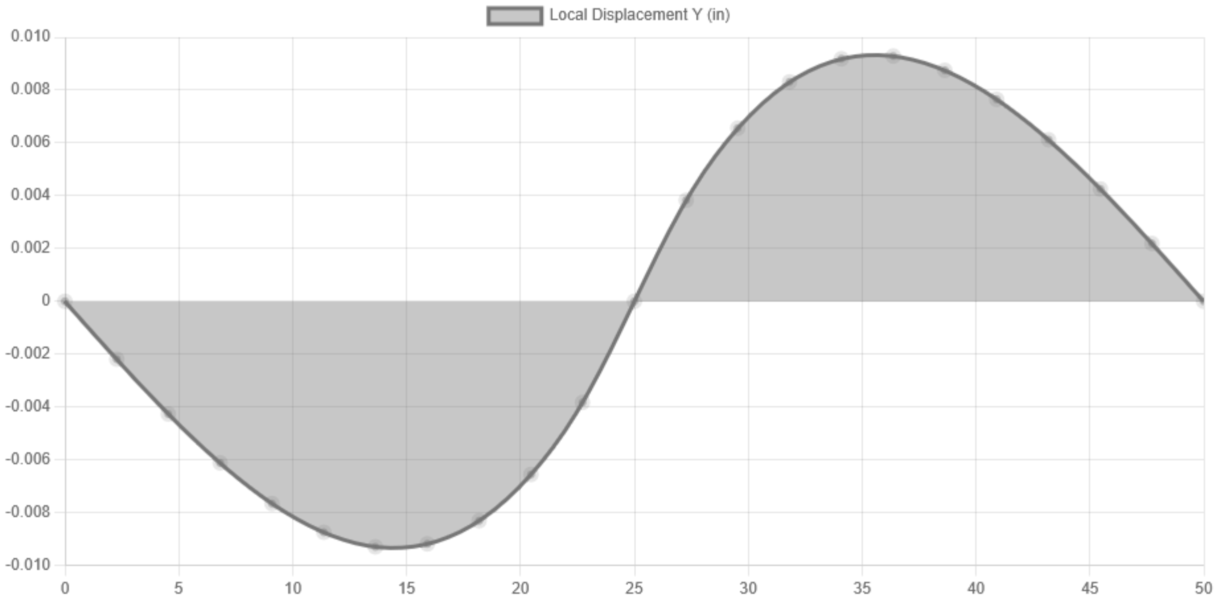
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	13.636	25	36.364	50
Total Deflection (in)	0 ?	0.009 ?	0 ?	0.009 ?	0 ?
Deflection/Span	-	L/64583	-	L/64583	-

The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the **model settings.**